

Solar C appendix

Sun · C · downloadable appendix

C.json v1.19 · 17 visible products · 0 quarantined entries

products

Independent admitted products; no combined headline abundance.

Highlighted Products

No adopted highlighted product has been selected. See the independent products below.

Error-bar Forest Plot

VIS — harps_solar_harps_molecfits_corrected A(C) dex; axis 8.31-8.71		
SET-AGSS21 · C I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied Reference Grade · ALL · n=3		8.469 ±0.013 stat ±0.041 syst
indicator not supplied · C I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied Codex Grade · ALL · n=5		8.536 ±0.125 stat ±0.145 syst
MOL-CH_Gband · C I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied Codex Grade · ALL · n=184		8.488 ±0.005 stat ±0.030 syst
ATOM-Cl_5052 · C I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied Codex Grade · ALL · n=1		8.476 ±0.091 stat ±0.030 syst
ATOM-Cl_5380 · C I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied Codex Grade · ALL · n=1		8.530 ±0.098 stat ±0.030 syst
MOL-C2_Swan · C I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied Codex Grade · ALL · n=167		8.449 ±0.007 stat ±0.030 syst
VIS — kpno_solar_atlas_solar_kpno_kurucz2005_corrected A(C) dex; axis 8.31-8.71		
SET-AGSS21 · C I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied Reference Grade · ALL · n=3		8.477 ±0.034 stat ±0.041 syst
indicator not supplied · C I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied Codex Grade · ALL · n=5		8.480 ±0.073 stat ±0.145 syst
VIS — kpno_solar_atlas_solar_kpno_molecfits_corrected A(C) dex; axis 8.31-8.71		
SET-AGSS21 · C I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied Reference Grade · ALL · n=3		8.456 ±0.037 stat ±0.041 syst
indicator not supplied · C I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied Codex Grade · ALL · n=5		8.503 ±0.077 stat ±0.145 syst

VIS — iag_fts_solar_atlas solar_iag A(C) dex; axis 8.31-8.71

indicator not supplied · C I · Synth · 1D-LTE · SYNTH ·
1D-LTE · line set not supplied
Codex Grade · ALL · n=2



8.473
±0.043 stat ±0.041 syst

VIS axis: 8.31 · 8.41 · 8.51 · 8.61 · 8.71 A(C) dex

red-optical — iag_fts_solar_atlas solar_iag A(C) dex; axis 8.48-9.21

SET-AGSS21 · C I · Synth · 1D-LTE · SYNTH · 1D-LTE ·
line set not supplied
Reference Grade · ALL · n=10



8.564
±0.024 stat ±0.050 syst

indicator not supplied · C I · Synth · 1D-LTE · SYNTH ·
1D-LTE · line set not supplied
Codex Grade · ALL · n=6



8.587
±0.024 stat ±0.050 syst

red-optical — kpno_solar_atlas solar_kpno_kurucz2005_corrected A(C) dex; axis 8.48-9.21

SET-AGSS21 · C I · Synth · 1D-LTE · SYNTH · 1D-LTE ·
line set not supplied
Reference Grade · ALL · n=10



8.624
±0.029 stat ±0.050 syst

indicator not supplied · C I · Synth · 1D-LTE · SYNTH ·
1D-LTE · line set not supplied
Codex Grade · ALL · n=6



8.628
±0.017 stat ±0.050 syst

red-optical — kpno_solar_atlas solar_kpno_molecfit_corrected A(C) dex; axis 8.48-9.21

SET-AGSS21 · C I · Synth · 1D-LTE · SYNTH · 1D-LTE ·
line set not supplied
Reference Grade · ALL · n=10



8.793
±0.178 stat ±0.050 syst

indicator not supplied · C I · Synth · 1D-LTE · SYNTH ·
1D-LTE · line set not supplied
Codex Grade · ALL · n=6



8.981
±0.201 stat ±0.050 syst

red-optical axis: 8.48 · 8.67 · 8.85 · 9.03 · 9.21 A(C) dex

How to read this forest. Each row is one independent product, preserving its band, holding, engine, indicator and line set. Engines and indicator families are never averaged. Solid bars show the supplied statistical uncertainty; wireframes show the supplied systematic uncertainty. The statistical basis and any incomplete error budget remain attached to the row. Grades describe the supplied line-pool provenance, not engine quality; evaluated theoretical CNO oscillator strengths are not relabelled as laboratory evidence. Quarantined, failed and unconstrained fits have no abundance mark. Experimental products remain explicitly not adopted. Literature references provide context, never tuning targets.

Solar carbon context

Carbon is traced by atomic C I, including forbidden [C I], and molecular CH, C₂ and CO indicators. Each probes different line formation and molecular-equilibrium sensitivities. Indicator families and atmospheric treatments remain separate products.

[C/O indicator strategy](#) · [Methodology](#)

Product provenance and caveats

SET-AGSS21 · C I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied: {"display": "Synth · 1D-LTE", "dominant_term": "gf scale (cited NIST class)", "gf_pool_graded": true, "gf_rung_summary": "gf rung 2 (gf scale (cited NIST class))", "grade": "Reference Grade", "n_excluded": 0, "n_lines": 3, "provenance": {"artifact_mtime": "2026-09-12T22:01:12Z", "copied_to": "data/results/band_products/CI_4200_6908_harps_solar_harps_molecfite_corrected_SYNTH_SET-AGSS21_products.csv", "host": "Sirius", "ingested_at": "2026-09-12T23:16:42Z", "path": "/home/damienabraxas/scratch/rya1214/data/results/band_products/CI_4200_6908_harps_solar_harps_molecfite_corrected_SYNTH_SET-AGSS21_products.csv", "sha256": "406b879c6c68a967a5e80597ed91ff6cd41c7cf14b692a2dac009a3502908119"}, "sigma_stat": 0.0128, "sigma_syst": 0.0414, "stat_basis": "measured — RMS of the random terms, 0.01284 dex at n_lines=3"}

indicator not supplied · C I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied: {"display": "Synth · 1D-LTE", "dominant_term": "gf scale (cited NIST class)", "grade": "Codex Grade", "n_excluded": 1, "n_lines": 5, "provenance": {"artifact_mtime": "2026-09-12T22:05:34Z", "copied_to": "data/results/band_products/CI_4200_6908_harps_solar_harps_molecfite_corrected_SYNTH_products.csv", "host": "Sirius", "ingested_at": "2026-09-12T23:16:42Z", "path": "/home/damienabraxas/scratch/rya1214/data/results/band_products/CI_4200_6908_harps_solar_harps_molecfite_corrected_SYNTH_products.csv", "sha256": "75fb417ba45f05e445002f8585d9892099d789814b2f0a8a13ce44013a6c4e39"}, "sigma_stat": 0.125, "sigma_syst": 0.145, "stat_basis": "measured — RMS of the random terms, 0.12498 dex at n_lines=5"}

SET-AGSS21 · C I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied: {"display": "Synth · 1D-LTE", "dominant_term": "gf scale (cited NIST class)", "gf_pool_graded": true, "gf_rung_summary": "gf rung 2 (gf scale (cited NIST class))", "grade": "Reference Grade", "n_excluded": 0, "n_lines": 3, "provenance": {"artifact_mtime": "2026-09-12T21:46:44Z", "copied_to": "data/results/band_products/CI_4200_6910_kpno_solar_atlas_solar_kpno_kurucz2005_corrected_SYNTH_SET-AGSS21_products.csv", "host": "Sirius", "ingested_at": "2026-09-12T23:16:42Z", "path": "/home/damienabraxas/scratch/rya1214/data/results/band_products/CI_4200_6910_kpno_solar_atlas_solar_kpno_kurucz2005_corrected_SYNTH_SET-AGSS21_products.csv", "sha256": "4ec999d0b163fb5b1f0e5f89f64551c7d72ba612b475f53a6186dd111f1d476f"}, "sigma_stat": 0.0342, "sigma_syst": 0.0414, "stat_basis": "measured — RMS of the random terms, 0.03418 dex at n_lines=3"}

indicator not supplied · C I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied: {"display": "Synth · 1D-LTE", "dominant_term": "gf scale (cited NIST class)", "grade": "Codex Grade", "n_excluded": 1, "n_lines": 5, "provenance": {"artifact_mtime": "2026-09-12T19:28:21Z", "copied_to": "data/results/band_products/CI_4200_6910_kpno_solar_atlas_solar_kpno_kurucz2005_corrected_SYNTH_products.csv", "host": "Sirius", "ingested_at": "2026-09-12T23:16:43Z", "path": "/home/damienabraxas/scratch/rya1214/data/results/band_products/CI_4200_6910_kpno_solar_atlas_solar_kpno_kurucz2005_corrected_SYNTH_products.csv", "sha256": "bb6f5f428dcde1d0ecdd937d3a244662bd90850be89d81d5e67ff44db3b9951b"}, "sigma_stat": 0.0734, "sigma_syst": 0.145, "stat_basis": "measured — RMS of the random terms, 0.07340 dex at n_lines=5"}

SET-AGSS21 · C I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied: {"display": "Synth · 1D-LTE", "dominant_term": "gf scale (cited NIST class)", "gf_pool_graded": true, "gf_rung_summary": "gf rung 2 (gf scale (cited NIST class))", "grade": "Reference Grade", "n_excluded": 0, "n_lines": 3, "provenance": {"artifact_mtime": "2026-09-12T21:54:37Z", "copied_to": "data/results/band_products/CI_4200_6910_kpno_solar_atlas_solar_kpno_molecfite_corrected_SYNTH_SET-AGSS21_products.csv", "host": "Sirius", "ingested_at": "2026-09-12T23:16:43Z", "path": "/home/damienabraxas/scratch/rya1214/data/results/band_products/CI_4200_6910_kpno_solar_atlas_solar_kpno_molecfite_corrected_SYNTH_SET-AGSS21_products.csv", "sha256": "316afdb734e8eba00cfc0d868d1f0a7c589fa1b6bce6ceccf266d48efda354af"}, "sigma_stat": 0.0368, "sigma_syst": 0.0414, "stat_basis": "measured — RMS of the random terms, 0.03676 dex at n_lines=3"}

indicator not supplied · C I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied: {"display": "Synth · 1D-LTE", "dominant_term": "gf scale (cited NIST class)", "grade": "Codex Grade", "n_excluded": 1, "n_lines": 5, "provenance": {"artifact_mtime": "2026-09-12T19:27:42Z", "copied_to": "data/results/band_products/CI_4200_6910_kpno_solar_atlas_solar_kpno_molecfite_corrected_SYNTH_products.csv", "host": "Sirius", "ingested_at": "2026-09-12T23:16:43Z", "path": "/home/damienabraxas/scratch/rya1214/data/results/band_products/CI_4200_6910_kpno_solar_atlas_solar_kpno_molecfite_corrected_SYNTH_products.csv", "sha256": "794443758ee802655d4c0e39771e3f18bf16ecf141e34c57e814383673bd4a62"}, "sigma_stat": 0.0771, "sigma_syst": 0.145, "stat_basis": "measured — RMS of the random terms, 0.07707 dex at n_lines=5"}

indicator not supplied · C I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied: {"display": "Synth · 1D-LTE", "dominant_term": "line-to-line scatter", "grade": "Codex Grade", "n_excluded": 0, "n_lines": 2, "provenance": {"artifact_mtime": "2026-09-12T19:34:05Z", "copied_to": "data/results/band_products/CI_5002_6910_iag_fts_solar_atlas_solar_iag_SYNTH_products.csv", "host": "Sirius", "ingested_at": "2026-09-12T23:16:44Z", "path": "/home/damienabraxas/scratch/rya1214/data/results/band_products/CI_5002_6910_iag_fts_solar_atlas_solar_iag_SYNTH_products.csv", "sha256": "278fdf8c8cadef61d76845ace7277bb3f8ef072110945a9f0eb38212d70c3660"}, "sigma_stat": 0.0435, "sigma_syst": 0.0414, "stat_basis": "measured — RMS of the random terms, 0.04350 dex at n_lines=2"}

SET-AGSS21 · C I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied: {"display": "Synth · 1D-LTE", "dominant_term": "gf scale (cited NIST class)", "gf_pool_graded": true, "gf_rung_summary": "gf rung 2 (gf scale (cited NIST class))", "grade": "Reference Grade", "n_excluded": 0, "n_lines": 10, "provenance": {"artifact_mtime": "2026-09-12T21:53:23Z", "copied_to": "data/results/band_products/CI_6910_9199_iag_fts_solar_atlas_solar_iag_SYNTH_SET-AGSS21_products.csv", "host": "Sirius", "ingested_at": "2026-09-12T23:16:44Z", "path": "/home/damienabraxas/scratch/rya1214/data/results/band_products/CI_6910_9199_iag_fts_solar_atlas_solar_iag_SYNTH_SET-AGSS21_products.csv", "sha256": "1f998d9176ed417b574963610e744cf71d37b8202925925dc7c46385b92cfa3b"}, "sigma_stat": 0.0236, "sigma_syst": 0.0503, "stat_basis": "measured — RMS of the random terms, 0.02355 dex at n_lines=10"}

indicator not supplied · C I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied: {"display": "Synth · 1D-LTE", "dominant_term": "gf scale (cited NIST class)", "grade": "Codex Grade", "n_excluded": 0, "n_lines": 6, "provenance": {"artifact_mtime": "2026-09-12T20:04:25Z", "copied_to": "data/results/band_products/CI_6910_9199_iag_fts_solar_atlas_solar_iag_SYNTH_products.csv", "host": "Sirius", "ingested_at": "2026-09-12T23:16:44Z", "path": "/home/damienabraxas/scratch/rya1214/data/results/band_products/CI_6910_9199_iag_fts_solar_atlas_solar_iag_SYNTH_products.csv", "sha256": "351556c428ab9458bbe5ca076921f601dbe8f1d58d63468ab9f47704e1da27ee"}, "sigma_stat": 0.0241, "sigma_syst": 0.0497, "stat_basis": "measured — RMS of the random terms, 0.02406 dex at n_lines=6"}

SET-AGSS21 · C I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied: {"display": "Synth · 1D-LTE", "dominant_term": "gf scale (cited NIST class)", "gf_pool_graded": true, "gf_rung_summary": "gf rung 2 (gf scale (cited NIST class))", "grade": "Reference Grade", "n_excluded": 0, "n_lines": 10, "provenance": {"artifact_mtime": "2026-09-12T21:30:13Z", "copied_to": "data/results/band_products/CI_6910_9199_kpno_solar_atlas_solar_kpno_kurucz2005_corrected_SYNTH_SET-AGSS21_products.csv", "host": "Sirius", "ingested_at": "2026-09-12T23:16:45Z", "path": "/home/damienabraxas/scratch/rya1214/data/results/band_products/CI_6910_9199_kpno_solar_atlas_solar_kpno_kurucz2005_corrected_SYNTH_SET-AGSS21_products.csv", "sha256": "a06c4b48f5c2e02d963f23067c8ec6d4e96f93a7a9965eeaffcbb9d026044d6c"}, "sigma_stat": 0.0288, "sigma_syst": 0.0503, "stat_basis": "measured — RMS of the random terms, 0.02879 dex at n_lines=10"}

indicator not supplied · C I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied: {"display": "Synth · 1D-LTE", "dominant_term": "gf scale (cited NIST class)", "grade": "Codex Grade", "n_excluded": 0, "n_lines": 6, "provenance": {"artifact_mtime": "2026-09-12T19:55:04Z", "copied_to": "data/results/band_products/CI_6910_9199_kpno_solar_atlas_solar_kpno_kurucz2005_corrected_SYNTH_products.csv", "host": "Sirius", "ingested_at": "2026-09-12T23:16:45Z", "path": "/home/damienabraxas/scratch/rya1214/data/results/band_products/CI_6910_9199_kpno_solar_atlas_solar_kpno_kurucz2005_corrected_SYNTH_products.csv", "sha256": "b1306dd10f9e5d02d62693073dd1905f1238bb1c95b8a0e214a408ebdf7033bb"}, "sigma_stat": 0.0165, "sigma_syst": 0.0497, "stat_basis": "measured — RMS of the random terms, 0.01645 dex at n_lines=6"}

SET-AGSS21 · C I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied: {"display": "Synth · 1D-LTE", "dominant_term": "line-to-line scatter", "gf_pool_graded": true, "gf_rung_summary": "gf rung 2 (gf scale (cited NIST class))", "grade": "Reference Grade", "n_excluded": 0, "n_lines": 10, "provenance": {"artifact_mtime": "2026-09-12T21:38:50Z", "copied_to": "data/results/band_products/CI_6910_9199_kpno_solar_atlas_solar_kpno_molecfit_corrected_SYNTH_SET-AGSS21_products.csv", "host": "Sirius", "ingested_at": "2026-09-12T23:16:46Z", "path": "/home/damienabraxas/scratch/rya1214/data/results/band_products/CI_6910_9199_kpno_solar_atlas_solar_kpno_molecfit_corrected_SYNTH_SET-AGSS21_products.csv", "sha256": "44902d5c6f72ebc94ff719729b24f7d2dc942cfadf04104a962c001f5a20585f"}, "sigma_stat": 0.1784, "sigma_syst": 0.0503, "stat_basis": "measured — RMS of the random terms, 0.17843 dex at n_lines=10"}

indicator not supplied · C I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied: {"display": "Synth · 1D-LTE", "dominant_term": "line-to-line scatter", "grade": "Codex Grade", "n_excluded": 0, "n_lines": 6,

"provenance": {"artifact_mtime": "2026-09-12T19:54:55Z", "copied_to": "data/results/band_products/CI_6910_9199_kpno_solar_atlas_solar_kpno_molecfi_t_corrected_SYNT_H_products.csv", "host": "Sirius", "ingested_at": "2026-09-12T23:16:46Z", "path": "/home/damienabraxas/scratch/rya1214/data/results/band_products/CI_6910_9199_kpno_solar_atlas_solar_kpno_molecfi_t_corrected_SYNT_H_products.csv", "sha256": "6fef78a4b4be1a60cca81c0ed3aa3ed47ebdff4b19f354c8080851002c100da7"}, "sigma_stat": 0.2009, "sigma_syst": 0.0497, "stat_basis": "measured — RMS of the random terms, 0.20086 dex at n_lines=6"}

MOL-CH_Gband · C I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied: {"display": "Synth · 1D-LTE", "dominant_term": null, "grade": "Codex Grade", "n_excluded": 0, "n_lines": 184, "provenance": {"artifact_mtime": "2026-09-13T04:18:40Z", "copied_to": "data/results/band_products/CI_MOL-CH_Gband_harps_solar_harps_molecfi_t_corrected_SYNT_H_MOL-CH_Gband_products.csv", "host": "Sirius", "ingested_at": "2026-09-13T04:21:44Z", "path": "/home/damienabraxas/scratch/rya1214/data/results/band_products/CI_MOL-CH_Gband_harps_solar_harps_molecfi_t_corrected_SYNT_H_MOL-CH_Gband_products.csv", "sha256": "3cb103caf07cc8ec6509d19061ecd8105bc7e19dcbb78c2aa50b4b0a927193ed"}, "sigma_stat": 0.005, "sigma_syst": 0.03, "stat_basis": "measured — 1 sigma from the chi2 curvature of THIS band's fit (pipeline.fit_constraint.curvature_sigma, rescaled to red_chi2 = 1); NaN where the curvature was not measurable, never 0 and never the retired 1.000 clip"}

ATOM-Cl_5052 · C I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied: {"display": "Synth · 1D-LTE", "dominant_term": null, "grade": "Codex Grade", "n_excluded": 0, "n_lines": 1, "provenance": {"artifact_mtime": "2026-09-13T04:18:40Z", "copied_to": "data/results/band_products/CI_ATOM-Cl_5052_harps_solar_harps_molecfi_t_corrected_SYNT_H_ATOM-Cl_5052_products.csv", "host": "Sirius", "ingested_at": "2026-09-13T04:21:44Z", "path": "/home/damienabraxas/scratch/rya1214/data/results/band_products/CI_ATOM-Cl_5052_harps_solar_harps_molecfi_t_corrected_SYNT_H_ATOM-Cl_5052_products.csv", "sha256": "054e5b65b4dd1e76e2b76dbd0000c36ce0afb0949cb0bc407906acb3b2cd5fe6"}, "sigma_stat": 0.091, "sigma_syst": 0.03, "stat_basis": "measured — 1 sigma from the chi2 curvature of THIS band's fit (pipeline.fit_constraint.curvature_sigma, rescaled to red_chi2 = 1); NaN where the curvature was not measurable, never 0 and never the retired 1.000 clip"}

ATOM-Cl_5380 · C I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied: {"display": "Synth · 1D-LTE", "dominant_term": null, "grade": "Codex Grade", "n_excluded": 0, "n_lines": 1, "provenance": {"artifact_mtime": "2026-09-13T04:18:40Z", "copied_to": "data/results/band_products/CI_ATOM-Cl_5380_harps_solar_harps_molecfi_t_corrected_SYNT_H_ATOM-Cl_5380_products.csv", "host": "Sirius", "ingested_at": "2026-09-13T04:21:44Z", "path": "/home/damienabraxas/scratch/rya1214/data/results/band_products/CI_ATOM-Cl_5380_harps_solar_harps_molecfi_t_corrected_SYNT_H_ATOM-Cl_5380_products.csv", "sha256": "a8246d3119d7a18e417c0d6939e27be45b86cb3952b3f6136c60046023b34c2c"}, "sigma_stat": 0.098, "sigma_syst": 0.03, "stat_basis": "measured — 1 sigma from the chi2 curvature of THIS band's fit (pipeline.fit_constraint.curvature_sigma, rescaled to red_chi2 = 1); NaN where the curvature was not measurable, never 0 and never the retired 1.000 clip"}

MOL-C2_Swan · C I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied: {"display": "Synth · 1D-LTE", "dominant_term": null, "grade": "Codex Grade", "n_excluded": 0, "n_lines": 167, "provenance": {"artifact_mtime": "2026-09-13T04:18:40Z", "copied_to": "data/results/band_products/CI_MOL-C2_Swan_harps_solar_harps_molecfi_t_corrected_SYNT_H_MOL-C2_Swan_products.csv", "host": "Sirius", "ingested_at": "2026-09-13T04:21:45Z", "path": "/home/damienabraxas/scratch/rya1214/data/results/band_products/CI_MOL-C2_Swan_harps_solar_harps_molecfi_t_corrected_SYNT_H_MOL-C2_Swan_products.csv", "sha256": "1d8e679c1fa71316ef24f80cf3e14e632e98cdf20f92ab40c80cbbb9c57cb33f"}, "sigma_stat": 0.007, "sigma_syst": 0.03, "stat_basis": "measured — 1 sigma from the chi2 curvature of THIS band's fit (pipeline.fit_constraint.curvature_sigma, rescaled to red_chi2 = 1); NaN where the curvature was not measurable, never 0 and never the retired 1.000 clip"}

Product visibility audit

Every feed entry is accounted for. Quarantined and archived entries retain their source disposition; their numeric fit outputs are available only in the labelled source download.

Withheld entries and source reasons (0)

Download the evidence

[Download PDF — Solar C appendix](#)

[Source feed — includes quarantined fit outputs, not adopted abundances](#)

[Complete product visibility audit](#)

[Shared website/PDF report](#)

[Reproducibility manifest](#)

[Complete references and product-to-source bindings](#)

Reproducibility

Generated from the pinned CNO campaign snapshot. Source commit

22ea65678043e016b12fb0defa596c10a200df88

Feed C.json v1.19 · updated 2026-09-25T19:39:46Z

Generated 2026-09-25T21:53:06-06:00 (source commit timestamp)

Feed SHA-256 44027d58c1340f25c31852f4440a2c37f774d1524539347da8ab399fc18a5f88

Generator scripts/generate_cno_publication.py v1.0.0

[Pinned scientific source](#)

References / Data & Model Sources

Sources are separated by their role. Registered grid lineage and quarantined diagnostics do not imply an adopted abundance or an applied correction.

See the source notes for the distinction between reference lineage and the treatment actually recorded in each feed entry.

Science and provenance source notes

Solar abundance and indicator analyses. Solar C atomic/molecular indicator lineage and atmospheric sensitivity; comparison literature, not an adopted Codex abundance. Sources: [1], [2].

Atomic data and gf authorities. The recorded C I/II pedigree is the partial NIST MCHF update described by Wiese & Fuhr; evaluated theory is not primary laboratory evidence. Sources: [3], [4].

Atomic data and gf authorities. AGSS21 names Li et al. for its C I gf lineage. This reference calculation is distinct from the campaign's NIST/MCHF adjudication; no gf source is substituted here. Sources: [5].

Model and grid sources. Registered C I line/grid lineage: Amarsi, Nissen & Skuladottir 2019, A&A 630, A104 — line-by-line 3D-NLTE / 1D-NLTE grids (CDS J/A+A/630/A104). The current feed treatment remains authoritative; a grid reference does not imply a correction was applied. Sources: [6].

Atomic data and gf authorities. Atomic C I entry; source-family pedigree only, not a substitute for a per-transition gf join. Sources: [3], [4].

Model and grid sources. Feed treatment: 1D-LTE; no NLTE/3D correction is claimed for this entry. Turbospectrum is the synthesis implementation. Sources: [7].

Molecular line data and delivery. CH line-data delivery: CH A-X: Masseron et al. 2014 (MNRAS 442, 1489) / 2022 update. Masseron compilation, VALD-based, distributed with iSpec/Turbospectrum (Gerber et al. 2023, A&A 669, A43; uploaded 2022 to keeper.mpdl.mpg.de, split per wavelength region) — see molecules-dir README.md The primary intake identifies Masseron 2014 as A&A 571 A47; the older delivery manifest's MNRAS locator conflicts with that record. Sources: [2], [7], [8].

Molecular line data and delivery. C2 line-data delivery: C2 Swan d³Pg–a³Pu (Brooke et al. 2013 / Masseron compilation). Masseron compilation, VALD-based, distributed with iSpec/Turbospectrum (Gerber et al. 2023, A&A 669, A43; uploaded 2022 to keeper.mpdl.mpg.de, split per wavelength region) — see molecules-dir README.md Sources: [2], [7], [9].

[Product-to-source index](#)

C-products-d336fe3e3260 · products · VIS · solar_harps_molecfits_corrected · SET-AGSS21 · 1D-LTE · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[7\]](#)

C-products-41dc4d995d0a · products · VIS · solar_harps_molecfits_corrected · not supplied · 1D-LTE · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[7\]](#)

C-products-d7f7f2d3feb3 · products · VIS · solar_kpno_kurucz2005_corrected · SET-AGSS21 · 1D-LTE · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[7\]](#)

C-products-639e603cf7be · products · VIS · solar_kpno_kurucz2005_corrected · not supplied · 1D-LTE · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[7\]](#)

C-products-2aa4ddfc06a1 · products · VIS · solar_kpno_molecfits_corrected · SET-AGSS21 · 1D-LTE · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[7\]](#)

C-products-9b3b9069e4e7 · products · VIS · solar_kpno_molecfits_corrected · not supplied · 1D-LTE · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[7\]](#)

C-products-3bc9c9cad7eb · products · VIS · solar_iag · not supplied · 1D-LTE · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[7\]](#)

C-products-355831cbb1ba · products · red-optical · solar_iag · SET-AGSS21 · 1D-LTE · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[7\]](#)

C-products-4fbb94c006f5 · products · red-optical · solar_iag · not supplied · 1D-LTE · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[7\]](#)

C-products-fcfcfc6a7f1 · products · red-optical · solar_kpno_kurucz2005_corrected · SET-AGSS21 · 1D-LTE · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[7\]](#)

C-products-7dc0d38da697 · products · red-optical · solar_kpno_kurucz2005_corrected · not supplied · 1D-LTE · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[7\]](#)

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C-products-886af3241708 · products · VIS · solar_harps_molecfits_corrected · ATOM-CI_5052 · 1D-LTE · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[7\]](#)

C-products-801532fb2d97 · products · VIS · solar_harps_molecfits_corrected · ATOM-CI_5380 · 1D-LTE · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[7\]](#)

C-products-1c1b6ac111e3 · products · VIS · solar_harps_molecfits_corrected · MOL-C2_Swan · 1D-LTE · not supplied · [\[1\]](#), [\[2\]](#), [\[7\]](#), [\[9\]](#)

Complete applicable bibliography

Solar abundance and indicator analyses

[1] — Asplund, M.; Amarsi, A. M.; Grevesse, N. (2021). The chemical make-up of the Sun: A 2020 vision. *A&A* 653, A141 DOI: 10.1051/0004-6361/202140445. [source record](#)

[2] — Amarsi et al. 2021, *A&A* 656 A113 DOI: 10.1051/0004-6361/202141384. [source record](#)

Atomic data and gf authorities

[3] — Kramida, A.; Ralchenko, Yu.; Reader, J.; NIST ASD Team (2024). NIST Atomic Spectra Database (version 5.11/5.12). National Institute of Standards and Technology, Gaithersburg, MD DOI: 10.18434/T4W30F. [source record](#)

[4] — Wiese, W. L.; Fuhr, J. R. (2006). New Critical Compilations of Atomic Transition Probabilities for Neutral and Singly Ionized Carbon, Nitrogen, and Iron. Proceedings of the NASA Laboratory Astrophysics Workshop (14-16 February 2006, UNLV, Las Vegas) [source record](#)

[5] — Li, Amarsi, Papoulia, Ekman & Jonsson 2021, MNRAS 502, 3780 DOI: 10.1093/mnras/stab214. [source record](#)

Model and grid sources

[6] — Amarsi, A. M.; Nissen, P. E.; Skuladottir, A. (2019). Carbon, oxygen, and iron abundances in disk and halo stars. A&A 630, A104 DOI: 10.1051/0004-6361/201936265. [source record](#)

[7] — Gerber, J. M.; Magg, E.; Plez, B.; Bergemann, M.; Heiter, U.; Olander, T.; Hoppe, R. (2023). Non-LTE radiative transfer with Turbospectrum. A&A 669, A43 DOI: 10.1051/0004-6361/202243673. [source record](#)

Molecular line data and delivery

[8] — Masseron et al. 2014, A&A 571 A47 DOI: 10.1051/0004-6361/201423956. [source record](#)

[9] — Brooke et al. 2013, JQSRT 124, 11 DOI: 10.1016/j.jqsrt.2013.02.025. [source record](#)